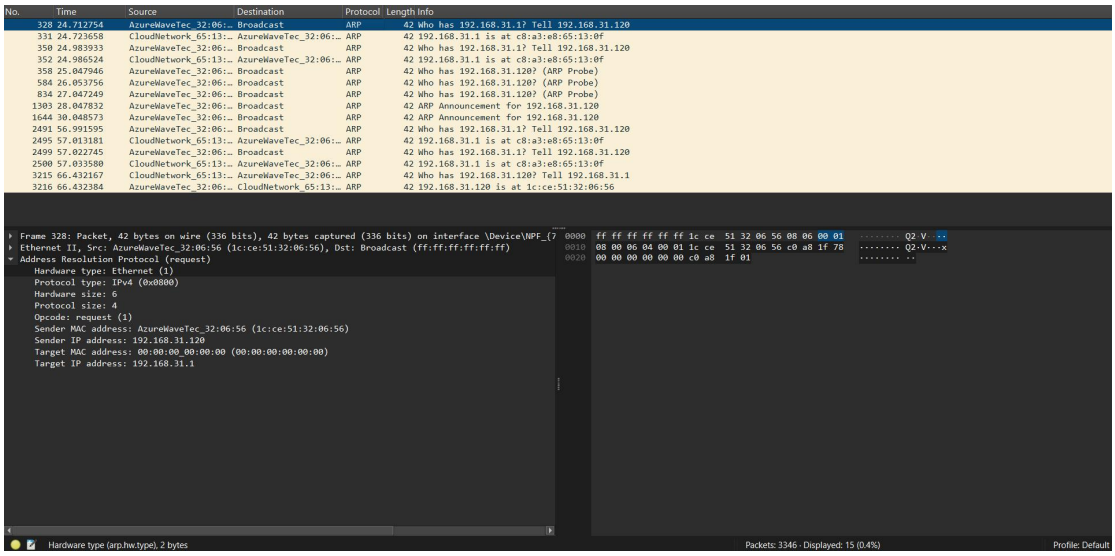
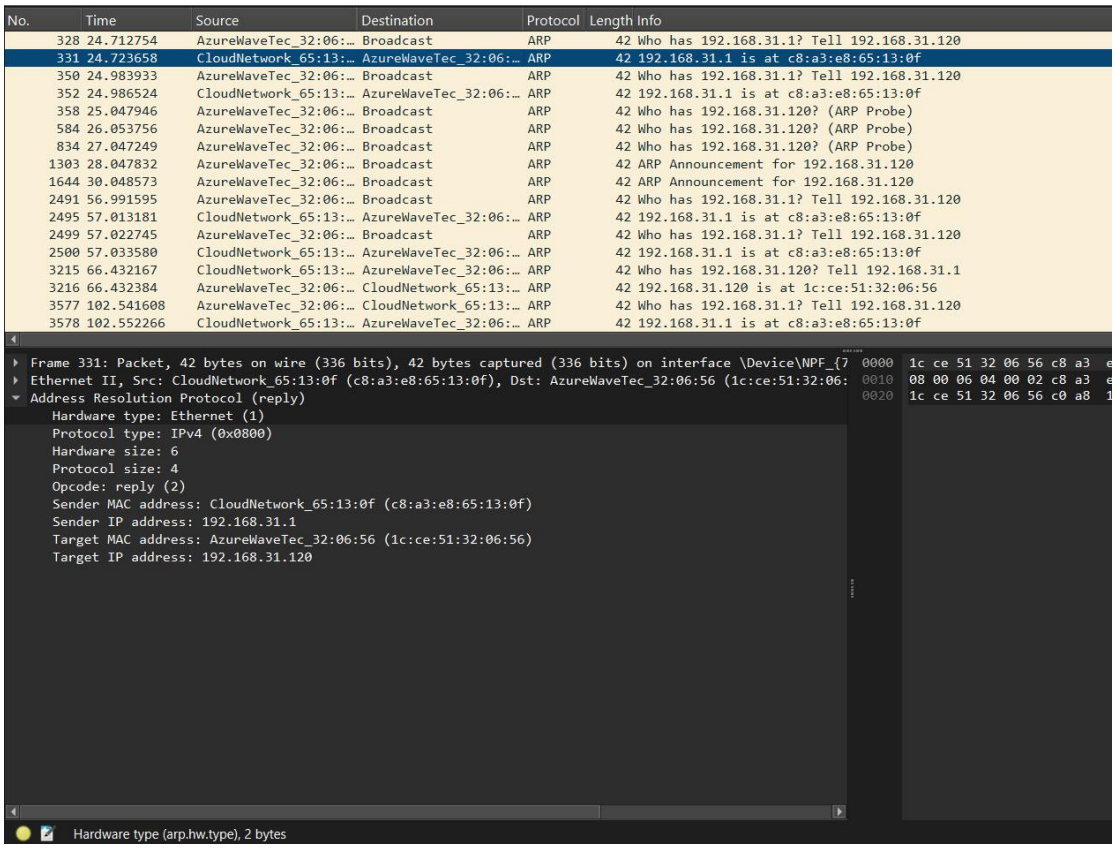
1)





- **ARP Request** is a broadcast message sent by a device asking *"Who has this IP address?"* when it needs the MAC address of another device (e.g., router).
- **ARP Reply** is a unicast response from the target device saying *"This IP is mine, here is my MAC address."*
- This process is essential because it maps **IP → MAC**, enabling actual packet delivery in a local network.

2)

10.0.0.0/24 , 4 Subset

$2^n = 4$

$\boxed{n = 2}$

/24 → /24+2 = /26

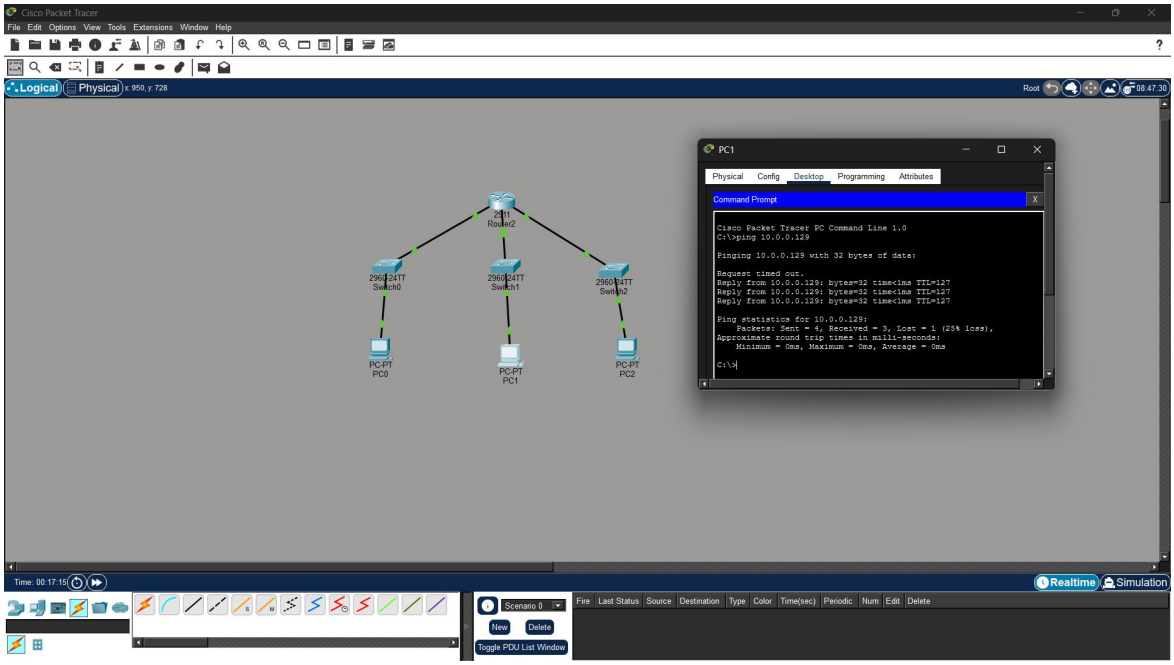
∴ Subnet = 255.255.255.192

Total Possible Subnets = /32

Available = $(2^{32-26}) = 2^6 = 64$

N/w ID	IP-Range	Broadcast
10.0.0.0/26	10.0.0.01 – 10.0.0.62	10.0.0.63
10.0.0.64/26	10.0.0.65 – 10.0.0.126	10.0.0.127
10.0.0.128/26	10.0.0.129 – 10.0.0.190	10.0.0.191
10.0.0.192/26	10.0.0.193 – 10.0.0.254	10.0.0.255

3)



4)

IP Address	Class	Address Range Check	Type
192.168.10.5	Class C	192.0.0.0 – 223.255.255.255	Private
172.20.15.1	Class B	128.0.0.0 – 191.255.255.255	Private
8.8.8.8	Class A	1.0.0.0 – 126.255.255.255	Public

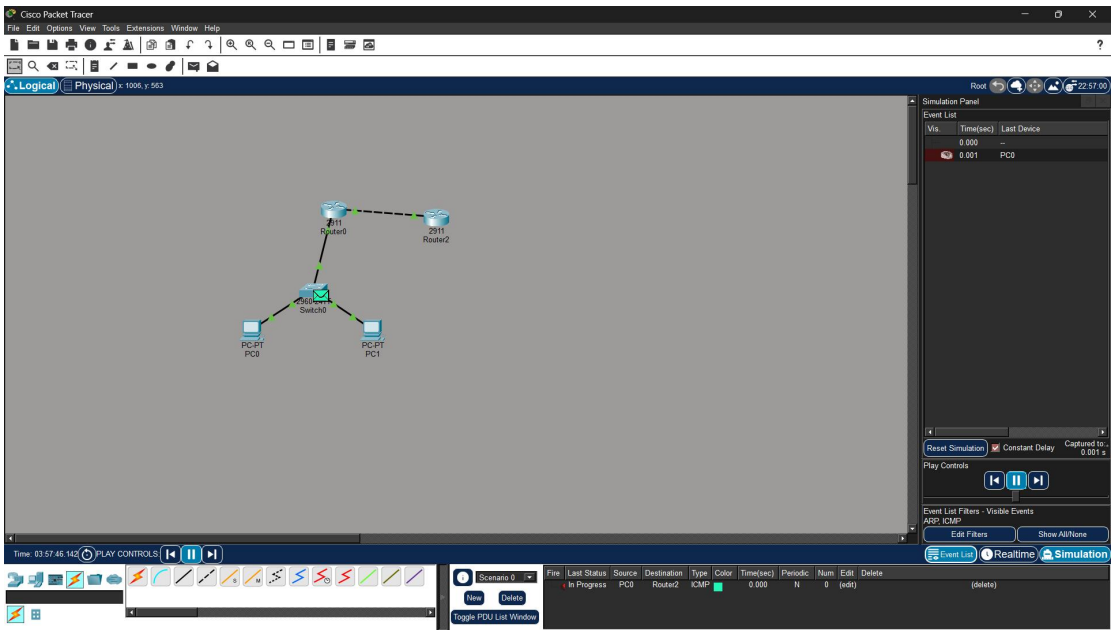
192.168.10.5 → Falls within 192.168.0.0 – 192.168.255.255 → Reserved for private networks

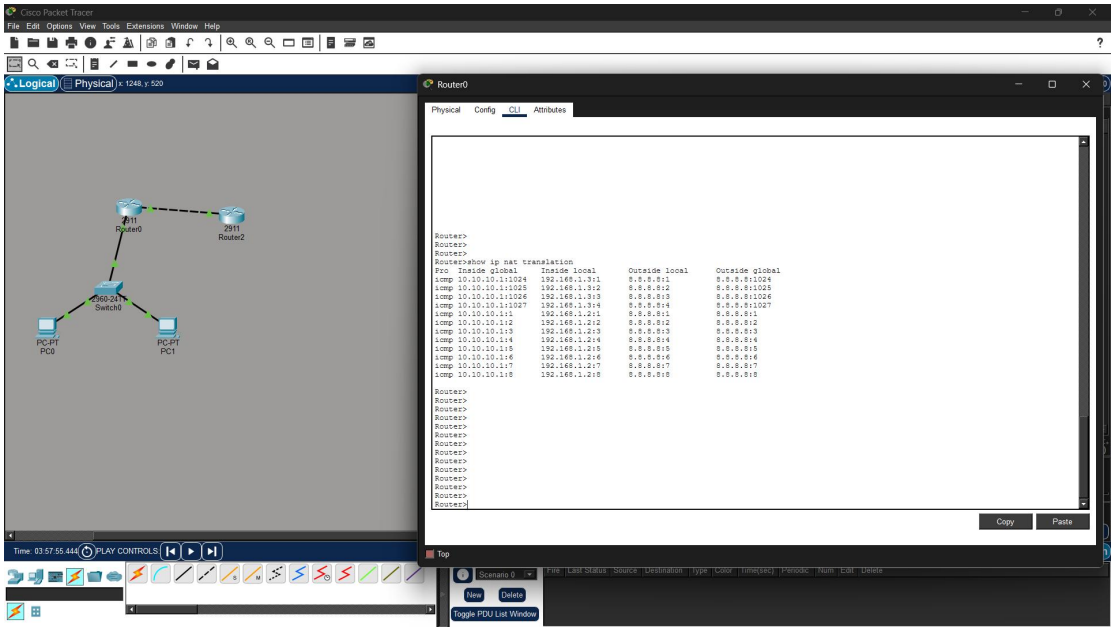
172.20.15.1 → Falls within 172.16.0.0 – 172.31.255.255 → Private IP range

8.8.8.8 → Does not fall under any private IP range → Public IP

Network Address Translation (NAT) is a technique used in networking where a router or gateway modifies IP address information in packet headers while they are in transit, allowing devices in a private network (using private IP addresses) to communicate with external networks like the internet using a single public IP address. When a device sends data, NAT replaces the private source IP with the router's public IP and maintains a mapping table to track connections; when responses return, the router uses this table to forward packets back to the correct internal device. This process conserves public IP addresses and adds a layer of security by hiding internal network details.

5)





The NAT translation table shows that internal devices 192.168.1.2 and 192.168.1.3 are communicating with the external IP 8.8.8.8. Their private IP addresses are translated to a single public IP 10.10.10.1 using different port numbers.

For example, traffic from 192.168.1.3 is translated as 10.10.10.1:1024, while traffic from 192.168.1.2 is translated as 10.10.10.1:1. This allows both devices to share the same public IP while maintaining separate sessions.